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Review article

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Heat of the night: sleep disturbance activates inflammatory mechanisms and induces pain in rheumatoid arthritis

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Abstract

Sleep has a homeostatic role in the regulation of the immune system and serves to constrain activation of inflammatory signalling and expression of cellular inflammation. In patients with rheumatoid arthritis (RA), a misaligned inflammatory profile induces a dysregulation of sleep–wake activity, which leads to excessive inflammation and the induction of increased sensitivity to pain. Given that multiple biological mechanisms contribute to sleep disturbances (such as insomnia), and that the central nervous system communicates with the innate immune system via neuroendocrine and neural effector pathways, potential exists to develop prevention opportunities to mitigate the risk of insomnia in RA. Furthermore, understanding these risk mechanisms might inform additional insomnia treatment strategies directed towards steering and reducing the magnitude of the inflammatory response, which together could influence outcomes of pain and disease activity in RA.

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Key points

- Patients with rheumatoid arthritis (RA), a chronic inflammatory disorder, frequently complain of symptoms of insomnia and pain.
- Inflammation can induce sleep disturbance, and in turn sleep disturbance increases inflammation in a feedforward loop, which is sustained by sympathetic arousal mechanisms and a downregulation of glucocorticoid receptor sensitivity.
- Sleep disturbance can also increase pain sensitivity, as mediated by increases in inflammation; in patients with RA, sleep loss induces heightened pain responses and activation of arthritis-related joint pain.
- Interventions that treat insomnia have the potential to interrupt the effects of sleep disturbance on inflammation, pain sensitivity and symptomatic progression in patients with RA.

Introduction

Rheumatoid arthritis (RA) is characterized by often symmetric, polyarticular disease involving the small joints of the hands and feet. As a genetically and environmentally determined disease, RA is a systemic inflammatory disorder that is chronic and progressive. Multiple potential pathogenic inflammatory pathways are thought to lead to joint inflammation and destruction that is pannus driven. In the synovial tissue, complex crosstalk exists between T cells, B cells, fibroblasts, macrophages, dendritic cells and others, which is facilitated by locally produced cytokines and chemokines, mainly IL-1β, TNF, receptor activator of nuclear factor (NF)-κB ligand (RANKL) and IL-6. In turn, IL-6 and other hormone-like inflammatory factors stimulate systemic disease sequelae outside the joints, leading to pain and alterations of sleep (such as insomnia), as reviewed elsewhere[1,2].

Insomnia and pain are two of the most prominent behavioural symptoms in individuals with RA. Inflammation is implicated in the dysregulation of sleep–wake activity and induction of pain. In turn, sleep disruption activates inflammation, which together suggests a positive feedback spiral between sleep and inflammation. Hence, a conceptual framework is needed to explain why some individuals with RA are at risk of insomnia. Moreover, an urgent need exists to understand the biological mechanisms that might be targeted to mitigate progression of pain symptoms and disease activity in individuals with RA who have comorbid sleep problems.

In this Review, we present evidence that inflammatory and autonomic arousal mechanisms, driven in part by social adversities (such as life stress or interpersonal difficulties), contribute to a misalignment of sleep–wake activity, such that sleep disturbance leads to excessive increases in inflammation, as coordinated by the central nervous system and multiple neural and endocrine effector mechanisms. In turn, inflammation can lead to further disturbances of sleep and this feedforward loop is hypothesized to serve as an accelerator of RA symptoms, with robust evidence that sleep disruption leads to increases in pain sensitivity and RA disease activity, likely mediated by activation of inflammatory pathways. This Review focuses on insomnia (that is, difficulties falling asleep or maintaining sleep) as it is the most frequent sleep disturbance in patients with RA, as noted below. Finally, this Review considers the ability of pharmacological and behavioural interventions to harness regulation of sleep and inflammation, in which treatment of insomnia reverses inflammation with implications for the progression of RA.

Sleep characteristics and functions

Sleep is a behavioural state associated with reduced sensitivity to external stimuli and an increased arousal threshold[3], and characterized by reduced mental and physical activity, with decreases in muscle activity and interactions with the environment. Electroencephalography (EEG) and polysomnography (PSG) have demonstrated that brain activity varies along an arousal continuum (from fully awake to deep sleep) driven by two processes: a homeostatic process (Process S) that interacts with a process controlled by the circadian pacemaker (Process C). Process S drives sleep propensity and contributes to sleep onset, duration and length; process C modulates sleep timing by synchronizing sleep homeostatic mechanisms with the circadian system, as aligned with behaviours and physiological systems, along the 24-h sleep–wake cycle[4].

Assessment of sleep, including insomnia, can be performed using several methods, including PSG, actigraphy, diagnostic interview, sleep diary and self-report questionnaires. PSG uses EEG, electromyography and electro-oculography[5] to define three aspects of sleep: sleep continuity (that is, sleep latency, total sleep time, wake time after sleep onset, sleep efficiency), sleep architecture (amount of non-rapid eye movement (NREM) sleep, or sleep stages N1, N2, N3 or deep sleep), and amount of rapid eye movement (REM) sleep (including latency to the first REM period and REM density). Over the sleep period, sleep architecture dynamically changes, showing cyclic transition from NREM to REM sleep, with each cycle lasting about 100 min. NREM sleep predominates during the early part of the night, and REM sleep occurs mainly in the later night. Whereas sleep is dimensionally scored as a quantal behavioural state[6] with discrete scoring of the various stages of sleep, spectral analytic methods show a continuous shifting of mixed EEG frequencies to predominantly lower EEG frequencies in the transition from awake to NREM sleep and N3 sleep[7].

Sleep–wake activity can also be objectively estimated by sleep actigraphy (for example, by using an activity monitoring device worn on the wrist[5,8]), which, when coupled with a sleep diary, provides information on sleep duration and sleep maintenance (for example, waking after sleep onset and sleep efficiency). As noted below, few studies in patients with RA have used actigraphy to evaluate sleep.

Insomnia, the focus of this Review, is evaluated by questionnaire and interview assessment. The clinical diagnosis of insomnia disorder is defined by the Diagnostic and Statistical Manual of Mental Disorders, fifth edition, which characterizes the number, severity, frequency and duration of patient-reported insomnia complaints and daytime impairment[9]. The Insomnia Severity Index (a seven-item scale)[10] and the Pittsburgh Sleep Quality Index (a 19-item self-report questionnaire) are the two most widely used self-report instruments[5], both of which evaluate sleep difficulties (that is, quality, duration, night-time wakening and daytime dysfunction).

The varied functions of sleep include promoting homeostasis by regulating the immune system[11,12] and other physiological systems such as the hypothalamic–pituitary–adrenal axis (HPA) and the sympathetic nervous system (SNS). Data also show that sleep constrains inflammatory output[13], regulates haematopoiesis[14], preserves clonal diversity[13], and slows epigenetic ageing[15] and decay of the haematopoietic epigenome[13]. Additionally, sleep has a necessary role in supporting modifications in brain connectivity related to memory consolidation[16,17], cognitive resilience[18] and emotional regulation[19].

Thus, sleep protects against age-associated inflammatory disorders (including cardiovascular disease)[14], cancer and neurodegenerative disease (such as Alzheimer disease)[20,21].

Sleep disturbance in RA

Insomnia symptoms affect nearly one-quarter of adults in the USA, and >10% of the population fulfil diagnostic criteria for insomnia or have a clinical diagnosis of insomnia[22,23]. Insomnia is often persistent, lasting up to 3 years in nearly 50% of patients[24]. Females have a two-fold greater risk of insomnia than males[23]; evidence shows that adults >60 years of age have a rate of insomnia that approaches 50%, which overlaps with age-related increases in chronic medical conditions such as RA[20]. Few studies have evaluated risk of insomnia in relation to ethnicity, although African American people are estimated to be twice as likely to report poor sleep quality as white people, which might be related to experiences of discrimination or due to disparities in socioeconomic status[25].

This Review focuses on insomnia in RA because self-reported insomnia complaints and insomnia disorder show a prevalence of 50–75% in patients with RA, more than twice the rate in the general population[26,27]; insomnia is also reported to be elevated in community-dwelling older adults with RA independent of depression[28]. Furthermore, among the various sleep disorders reported in patients with RA, insomnia is the most prevalent, with a prevalence that is reported to be more than twice that of obstructive sleep apnoea (OSA)[27]. Indeed, a matched comparison study of patients with RA versus controls found no difference in the prevalence of OSA[29], although prospective data indicate a greater risk of developing OSA in patients with RA than in those without RA[30]. Moreover, disturbances of sleep duration and sleep maintenance, two of the most common symptoms of insomnia, are linked to inflammation, pain sensitivity and disease activity in patients with RA[31] (reviewed below). By contrast, OSA and its severity are not reported to be associated with disease activity in RA[27,32].

Despite the high frequency of insomnia and insomnia complaints, few controlled studies have quantitatively assessed sleep (using EEG) in patients with RA, although evidence suggests that these patients have longer sleep latency, poorer sleep efficiency, more awakenings and more alpha-EEG arousals during N3 or slow wave sleep than healthy controls[33–35]. To our knowledge, only two pilot studies have used actigraphy to assess the number of nocturnal awakenings in patients with RA[36,37]. Moreover, the majority of research is limited by a lack of comparison controls, small sample sizes[38–40] and reliance on traditional sleep scoring methods, with a relative absence of studies that provide an integrated assessment of sleep and neurobiological mechanisms (such as inflammation) in RA.

Mechanisms contributing to sleep disturbance in RA

Psychosocial and psychological mechanisms

Pathophysiological models of chronic insomnia frame persistent poor sleep quality in terms of diathesis-stress, such that predisposing factors including genetics and adverse early life experiences[41–43] contribute to elevated neurobiological states of hyperarousal (Fig. 1). Indeed, early adversity alters emotional regulation and arousal circuits, including salience networks, that predispose individuals to heightened emotional and physiological arousal responses. Such arousal is further maintained by maladaptive behavioural factors (for example, irregular sleep–wake activity and self-medication with alcohol to decrease arousal level)[44,45]. In RA, a dose–response relationship exists between early life stress and arthritis incidence[46,47], especially in females[48].

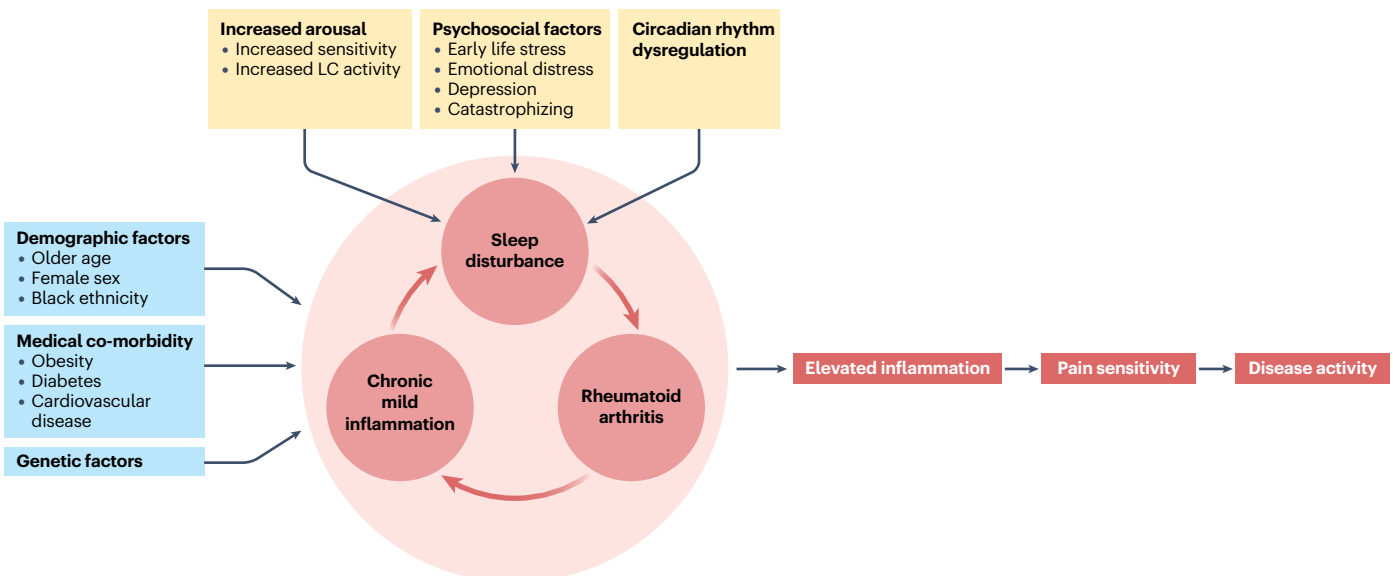

Fig. 1 | Model depicting the impact of sleep disturbance and inflammation on pain and disease activity in patients with RA. Predisposing factors such as demographic factors (older age, female sex and black ethnicity), medical co-morbidities (obesity, diabetes mellitus and cardiovascular disease) and genetic factors contribute to disturbances in sleep and chronic mild inflammation. Precipitating mechanisms including psychosocial factors (such as early life stress, depression and the cognitive style of catastrophizing) contribute to sleep disturbance directly and indirectly through increases in arousal, sensitivity to stimuli and activity of the locus coeruleus (LC), and through alterations in circadian rhythm. Together, these pre-disposing and precipitating factors drive a feedforward interaction between sleep disturbance and inflammation in patients with RA, which leads to further elevated levels of inflammation. When sleep disturbance and inflammation are perpetuated over time by the ongoing input from precipitating mechanisms, increases in symptoms of pain and pain sensitivity occur followed by progression of RA disease activity.

When exacerbated by stress, depression and pain catastrophizing reciprocally contribute to onset and perpetuation of sleep disturbance, which further drives pain, disease activity and disability in RA[49–51]. Pain catastrophizing involves pain-specific, cognitive-emotional processes that include the tendency to ruminate about pain, magnifying its threat, and related helpless feelings. Sleep disturbance, depression and pain catastrophizing reciprocally interact[52]. For example, poor sleep efficiency and high levels of pain catastrophizing predict heightened central pain processing and pain symptoms in patients with osteoarthritis[53] and RA[54], which suggests that these interacting constructs require routine assessment and/or targeted treatment.

Arousal mechanisms

As noted above, PSG demonstrates frequent arousals, awakenings and extremity movements in patients with RA[35,55], which is thought to be due to increased sensitivity to stimuli as a consequence of early life stress, acute and chronic emotional distress, mood disorders, depression and pain[56]. Furthermore, inflammatory load (for example, high levels of TNF) can drive arousal and sleep fragmentation[38,57]. Increased sensitivity to external and internal stimuli is also influenced by the central sympathetic noradrenergic system. Indeed, the efferent SNS is centrally integrated in the insular cortex, medial prefrontal cortex and nuclei of the hypothalamus[58], which are tightly coupled to the HPA axis. The locus coeruleus, which is coupled to the amygdala[58,59], strongly influences these limbic elements[58], with effects on sleep; total locus coeruleus quietness is necessary for normal REM sleep[58]. Given that REM sleep promotes neuronal plasticity and memory consolidation, and is a prerequisite for overcoming emotional stimuli and experiences during the day[45], an active locus coeruleus during REM sleep leads to hyperarousal, insomnia and disturbed emotional regulation[45].

The mechanisms that account for increases in locus coeruleus activity in RA are not fully understood. However, release of corticotrophin-releasing hormone (CRH), endogenous opioids and inflammatory states (for example, high levels of IL-1β)[60] can induce locus coeruleus neuronal discharge and peripheral SNS activity[60]. Other factors present in RA, including angiotensin II, NF-κB, reactive oxygen species, IL-6 and Toll-like receptor (TLR)4 activation, might contribute to locus coeruleus dysfunction and might increase peripheral SNS activity[61]. Baroreceptor dysfunction is common in patients with RA[62] and might also drive arousal.

Circadian rhythm dysregulation

Circadian rhythm is an important factor in the two-process model of sleep regulation[63]. Regulation of the circadian pacemaker in the superchiasmatic nucleus is altered[64] when elevated levels of inflammation occur[65,66], such as in RA. For example, higher peak values of IL-6 relative to cortisol in the early morning hours probably interfere with sleep quality[66–68]. Given that circadian rhythmicity is tightly linked to immune function[12], immune cell migration and immune cell precursor production in bone marrow[69,70], alterations in the circadian rhythm might stimulate a platform of disordered immunity[13,14,70]. Whether this new situation can stimulate autoimmunity in the first place is not yet known but has been discussed by some authors[71].

Inflammatory mechanisms

During times of inflammatory activation, the central nervous system (CNS) receives inflammatory signals via multiple pathways including neural innervation (vagal afferents), humoral or circulating mediators, and active mechanisms of blood–brain barrier transport (Box 1). In turn, neurotransmitters (such as serotonin and dopamine) are altered, with resultant changes in behaviour, which are often characterized as 'sickness behaviours' involving social withdrawal and changes in sleep[72,73].

In experimental models of infectious challenge, animals typically show increases in NREM sleep and decreases in REM sleep[74–76], with evidence that viral activation of pro-inflammatory cytokines (such as IL-1, IL-6, TNF and type I interferon) induces increases in N3 sleep in a cytokine-receptor-specific manner[77,78]. Pharmacological administration of an inflammatory cytokine results in alterations in sleep architecture, although effects differ in animals versus humans: increases in NREM sleep are induced in animals[75], whereas decreases in NREM sleep, N3 sleep and REM sleep are reported in humans in response to exogenous doses of IL-2, for example[79].

Endogenous levels of cytokines have a homeostatic role in the regulation of sleep, in which inflammatory cytokines act to drive sleep propensity, with similar effects in animals and humans. During the sleep period in animals, brain levels of IL-1 and TNF mRNA and protein are higher than in the awake period[80], and elevated production of these cytokines in response to sleep deprivation drives increases in recovery sleep via cytokine-dependent receptor mechanisms[75]. Studies in

Box 1

Pathways between peripheral inflammation and the CNS

Inflammatory signals communicate with the central nervous system (CNS) via molecular pathways, which can regulate sleep (reviewed elsewhere[75,76]). The afferent vagus nerve expresses inflammatory cytokine receptors. When activated by inflammatory cytokines such as IL-1 (ref. 73), the vagus nerve signals multiple brain regions, such as the nucleus of the solitary tract, the ventrolateral medulla, the paraventricular and supraoptic nuclei of the hypothalamus and the amygdala, all of which are involved in the regulation of sleep. Inflammatory cytokines can induce sleep, which can be abrogated by vagotomy[221].

The circumventricular organs and the choroid plexus of the CNS contain macrophage-like cells. TLR activation of these macrophages by peripheral inflammatory signals leads to the production of inflammatory cytokines that enter the CNS by volume diffusion[222]. In addition, activation of endothelial cells leads to prostaglandin E2 production that induces inflammatory activation in the CNS.

Distinct from passive diffusion, inflammatory molecules can be actively transported across the blood–brain barrier[223]. CNS processes, including sleep and circadian mechanisms, modulate the transport of inflammatory cytokines such as IL-1, IL-6 and TNF, with evidence that sleep patterns dynamically increase transport across the blood–brain barrier[224].

Finally, activated monocytes and lymphocytes traffic to the brain vasculature and brain parenchyma when inflammatory cytokines stimulate brain microglia to produce CCL2 (also known as MCP1). Inflammatory activation of astrocytes can also induce these cells to produce CCL2, further facilitating CNS attraction of immune cells[225].

humans are limited, although TLR4-stimulated monocyte production of IL-6 and TNF prior to sleep onset correlates with increases in sleep efficiency and N3 sleep[55,76].

By contrast, in patient populations that show excessive increases in inflammatory cytokines, beyond typical endogenous levels, evidence indicates that abnormal increases in TNF, for example, have opposing effects on sleep and induce alterations in normal sleep architecture. In people with inflammatory disorders such as RA who typically show excessive levels of inflammation, antagonism of TNF with the monoclonal antibody infliximab induces increases, not decreases, in relative amounts of N3 sleep[38] but has no effect on REM sleep. TNF antagonists were also found to improve sleep continuity (that is, sleep latency and sleep efficiency)[38,81] as well as sleep quality[82–85] in patients with RA, with similar findings following administration of an IL-6 antagonist[86] and the Janus kinase inhibitor, tofacitinib[87]. Findings generalize to other patients who similarly have inflammation, low amounts of N3 sleep and high amounts of REM sleep. For example, pharmacological neutralization of TNF induced a short-term (that is, one night) decrease of REM sleep in alcohol-dependent persons[88]. Despite the promise of these clinical and experimental findings, a prospective study of patients with RA did not demonstrate a significant association between initiation or change of biologic therapies and improvement in sleep disorders including insomnia[89]. Hence, although evidence from some clinical trials suggests an improvement in patient-reported sleep quality[84,87,90,91], DMARD therapies might not be enough to prevent or treat insomnia in patients with RA; interventions that target insomnia are needed even when patients are treated with potent biologic therapies that reduce inflammation.

Genetic factors

The heritability of insomnia ranges from 22 to 59%[92], with evidence that 956 gene variants are linked to insomnia, as found in a genome-wide association study (GWAS) of >1 million non-rheumatic adults[91]. Four overlapping genes are linked to RA and insomnia, each of which encodes proteins that have an immune-stimulating role: *PRRC2A* (also known as *BAT2;* involved in antigen presentation), *RASGRP* (activates the MAP kinase–ERK cascade and regulates T cell and B cell development), *LEMD2* (involved in cell signalling and differentiation) and *P2RX3* (encodes an ATP-receptor that has a role in pain perception).

Sleep and regulation of inflammation

Sleep has a dynamic role in the regulation of innate immunity, and normal sleep is associated with nocturnal increases in markers of inflammation (Box 2). However, this phasic coupling of innate immunity and sleep becomes misaligned when sleep is disrupted, leading to a shift in inflammatory activation from the night-time to the day. When sleep disturbance becomes chronic, systemic inflammation increases during the night and day, as reviewed elsewhere[11].

Using experimental models of sleep disruption (such as sleep deprivation, multiple night sleep restriction and sleep fragmentation), the causal link between sleep disruption and increases in markers of inflammation has been demonstrated. For example, sleep disruption induces the following: increases in circulating levels of C-reactive protein (CRP), IL-6 and TNF[93]; increases in ex vivo mitogen-stimulated production of pro-inflammatory cytokines; and increases in TLR4-stimulated and resting monocyte production of IL-6 and TNF[94]. Additionally, sleep disruption affects upstream inflammatory mechanisms by inducing increases in mRNA levels of IL-6 and TNF[94], and inducing activation of inflammatory transcriptional pathways including NF-κB[95] and the signal transducer and activators of transcription (STAT) family of proteins[96]. Further findings demonstrate that partial night sleep deprivation, or disruption of sleep during the first or second part of the night, leads to upregulation of a gene ensemble that includes the master circadian regulator, several 'immediate early genes' involved in coordinating cellular signal transduction, and multiple inflammatory response genes[94]. Moreover, those with pre-existing sleep disturbance, as well as females in general, demonstrate a heightened sensitivity to the effects of sleep loss on cellular and transcriptional inflammatory activation[97]; notably, these subgroups also have a heightened risk of inflammatory disorders including RA[20].

Despite substantial experimental and observational evidence in healthy adults, findings linking sleep to inflammation in RA are limited. In patients with RA, one observational study found that sleep disturbance correlated with an overnight increase in tenderness in the peripheral joints, possibly due to increases in inflammation[39]. Further, because activation of stress arousal mechanisms is thought in part to drive disturbances of sleep, it is possible that observations linking stress to regulation of inflammation in RA might also be relevant to sleep disturbance in these patients. To this point, in an experimental model of psychological stress, patients with RA had an exaggerated inflammatory response, which was mitigated by treatment with anti-TNF medications (such as infliximab, etanercept or adalimumab)[98]. Given that acute psychological stress is known to induce NF-κB activation[99,100], it is possible

Box 2

Sleep and the nocturnal profile of innate immunity

Sleep induces increases in innate immunity, which is thought to support the host immune response and prepare for infectious exposure or injury that might occur the following day. Such priming of the immune system allows for a rapid, and possibly more robust, response to infectious threats[11,226,227]. These pathways, and related effector mechanisms, have been previously reviewed and illustrated elsewhere[11,12,228].

Briefly, sleep, as opposed to circadian factors, leads to nocturnal increases in the activity of natural killer (NK) cells[229,230], as well as markers of systemic inflammation including IL-6 and TNF. Although there are circadian-dependent effects on IL-6 (for example, with peaks at 7 p.m. and at 5 a.m.[231]), sleep deprivation in the early part of the night delays nocturnal increases of IL-6 until sleep onset, and night-time levels of IL-6 are decreased during total night sleep deprivation[231,232]. In addition, nocturnal sleep leads to increases in the levels of the trans-signalling molecule, soluble IL-6 receptor (sIL-6R)[233], which are pronounced during the late part of the night. Likewise, stimulated production of IL-6 is elevated during the late part of the night, owing in part to increases in REM sleep[232]; periods of REM sleep, compared with N3 sleep, are associated with higher levels of IL-6 (ref. 232). Finally, sleep leads to increases in the stimulated production of TNF[234], although neither circulating levels of TNF, soluble receptor (sTNF-R), nor resting or unstimulated levels of monocyte expression of TNF[235] are altered by sleep (depending probably on rapid internalization of this cytokine).

that TNF antagonists block activation of NF-κB. Indeed, IκBα and IκBγ, known inhibitors of NF-κB, are upregulated in vitro by infliximab[101]. Other studies have found that patients with RA with high disease activity show greater increases of CRP levels 30 min after exposure to acute psychological stress than those with low disease activity[102]; however, a differential increase in LPS-stimulated production of IL-6 was not found in patients with RA exposed to stress[103].

Similar to experimental findings, naturalistic, observational studies have also found that sleep disturbance is associated with increases in circulating markers of inflammation. Our meta-analytic review found that reported sleep disturbance is associated with higher levels of CRP and IL-6 (ref. 93); the number of studies was too small to detect an effect of sleep disturbance on TNF and other cytokines. Likewise, shorter sleep duration as evaluated by subjective or objective methods was linked to higher CRP levels, but not IL-6 levels. Larger effect sizes were found in younger and female subgroups, with exploratory results also suggesting an increased risk of inflammation in African American populations[104], and for those with poor social relationships[105]. Variability in sleep duration is also associated with increases in CRP levels[106]. Finally, poor sleep quality correlates with increases in TLR4-stimulated monocyte production of IL-6 and TNF as well as elevated inflammatory transcriptional factor activity that correlates with transcription markers of SNS and HPA axis function[107]. Importantly, the effect size linking sleep disturbances to inflammation is comparable with that demonstrated for multiple demographic (such as age and race) and biobehavioural (such as BMI and physical activity) factors[93,108]. Finally, prospective data show that poor sleep quality predicts increases in inflammatory cytokines in females[109]; self-reported sleep disturbance and short sleep duration predict subsequent increases in CRP[110,111]; and objective measures of short sleep duration (<5 h) and sleep fragmentation (nocturnal wakening for >90 min) predict an increase in a composite measure of inflammation, which taken together mediate mortality risk[111,112].

In addition to evidence that sleep disruption alters the expression of immune cell products such as IL-6 and TNF, poor sleep quality and insomnia are associated with increases in numbers of blood myeloid cells in humans[113,114], possibly due to increases in haematopoietic stem progenitor cell cycling in the bone marrow, as demonstrated in mouse models[14]. Furthermore, enhanced haematopoietic activity persists even during recovery from sleep disruption, which seems to contribute to a persistent heightened number of blood monocytes[13]. Consistent with human findings that chronic sleep disturbance accelerates epigenetic ageing of leukocytes[15], sleep disruption in mouse models also compromises the capacity for lymphoid differentiation. Together, it seems that sleep disturbance drives epigenetic re-programming, which primes immune cells towards the development of an exaggerated inflammatory response to challenge[13]. In patients with RA, we hypothesize that sleep disturbance might initiate changes in haematopoietic clonal diversity and the epigenome, contributing to accelerated epigenetic ageing, increases in inflammatory output, along with promotion of RA disease activity and related inflammatory co-morbidities such as atherosclerosis, other cardiovascular disease, diabetes mellitus and obesity[14].

Sleep, neuroendocrine mechanisms and inflammation

HPA axis

During normal sleep, N3 sleep predominates during the first part of the night coincident with an absolute 24-h minimum of HPA axis activity[11], which is thought to be permissive for immune activity necessary to prime anti-infectious and memory immunity[12]. By contrast, when REM sleep occurs (mainly in the late part of the night) HPA activity becomes dominant, reaching its absolute maximum shortly after wakening. In patients with RA, levels of cortisol are reported to be normal (or high) during the day but levels of adrenal androgens are low[115], which is thought to lead to glucocorticoid resistance and perpetuation of inflammation (and associated dysregulation of sleep[11]) (Fig. 2).

Sympathetic nervous system

With sleep onset, efferent SNS activity is low[116,117], especially during N3 sleep[116,117]. During REM sleep, relative to N3 sleep, higher peripheral SNS activity is observed with increases in SNS neurotransmitters, sympathetic dominance in heart rate variability, and higher systolic and diastolic blood pressure[116,117]. In patients with RA, SNS activity is up to 30% higher[62,118], possibly because of baroreceptor dysfunction[62]. These findings suggest two major consequences[119]: sleep in patients with RA is disrupted owing to CNS hyperarousal, and peripheral inflammation is stimulated directly by sympathetic neuronal influences.

Several mechanisms are thought to underlie sympathetically stimulated peripheral inflammation: first, the SNS reaches primary and secondary lymphoid organs and similarly inflamed tissue through sympathetic nerve fibres[118]. Under conditions of high neurotransmitter concentrations of noradrenaline, some sympathetic influences are anti-inflammatory (for example, β2-adrenergically mediated inhibition of phagocytosis, neutrophils, NK cells, T helper 1 immunity, TNF, IL-2, IL-12, IFNγ), whereas other influences are proinflammatory (that is, immune cell mobilization, immune cell migration, chemotaxis, energy provision to an active immune system, T helper 2 immunity and others)[118]. Importantly, when inflammation is chronic, as in RA, anti-inflammatory pathways are blocked owing to a loss of sympathetic nerve fibres in inflamed secondary lymphoid organs and by a switch from anti-inflammatory to pro-inflammatory non-canonical receptor signalling of β-adrenoceptors[118]. Efferent SNS activity is also thought to promote metabolic activity (that is, the release of energy-rich fuels such as glucose, amino acids and lipids), which is necessary to support an inflammatory response for localized infections and wound healing.

Gonadal and other hormonal systems

Low serum levels of testosterone are associated with severe OSA and other disturbed measures of sleep[120]. Given that sleep apnoea can be present in RA[121], the well-known loss of adrenal and gonadal androgens in RA might have a role in this and other sleep problems[122]. Loss of these androgens is linked to a proinflammatory status, which in turn might contribute to sleep disorders. Oestrogen levels have been linked to better sleep[120], but oestrogen levels are not typically changed in patients with RA.

Consideration of age, sex and ethnic differences

During normal ageing, humans lose sex hormones around the age of 50, which is more striking in females than males, and serum levels of cortisol increase relative to sex hormones. The loss of oestrogen and progesterone in females can foster autoimmune diseases[123]. Similarly, the general loss of adrenal androgens supports inflammatory diseases in females and males[123], with effects on sleep as noted above.

Sexual dimorphism also applies to locus coeruleus function; females show increased locus coeruleus sensitivity to stressful events, which provides a molecular basis for a higher prevalence of

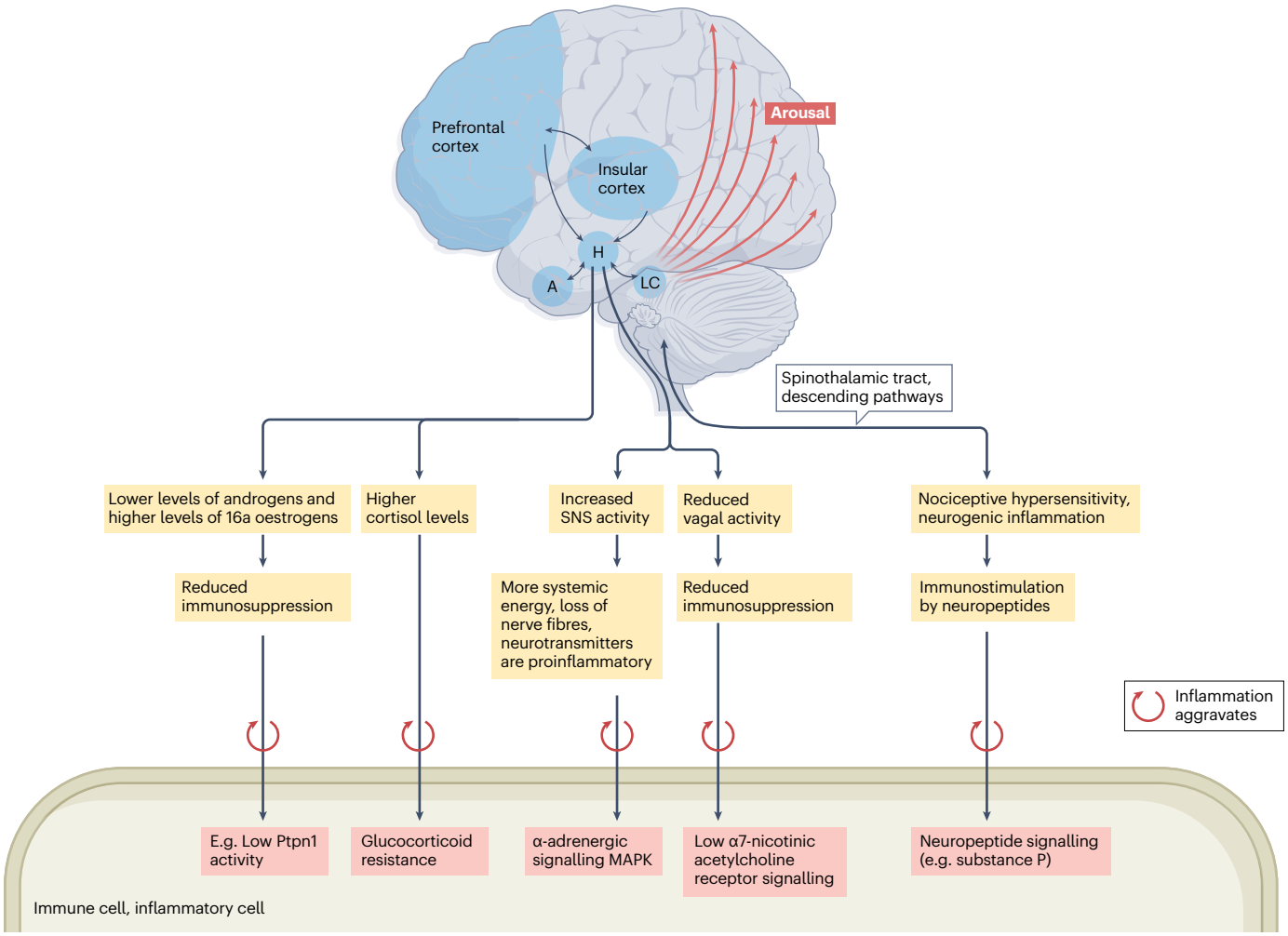

Fig. 2 | Schematic model of neural sites and effector mechanisms involved in sleep regulation of inflammation. Sleep disturbance and insomnia activate peripheral inflammation. Several brain centres are involved in coordinating effector mechanisms in response to sleep disturbance including the prefrontal cortex, hypothalamus (H), amygdala (A), insula and locus coeruleus (LC), which projects to the brain to alter arousal. Owing to the complex number of brain areas involved in arousal and sleep–wake activity, not all neural substrates are shown. Effector pathways from these various brain centres alter downstream neuroendocrine and neural pathways as shown, with changes in inflammatory activation as indicated for each pathway. Lower vagal activity has been described in RA[180]. Nociceptive hypersensitivity is typical in inflammatory arthritis as a consequence of peripheral, spinal and central nervous system alterations[140]. MAPK, MAP kinases; Ptpn1, protein tyrosine phosphatase non-receptor type 1; SNS, sympathetic nervous system.

stress-related diseases in females[58]. Locus coeruleus function has not been studied in RA, although females with RA might be more sensitive to arousal.

The bed nucleus of the stria terminalis, which is linked to anxiety and other social behaviours, also shows sexual dimorphism. Additionally, the bed nucleus of the stria terminalis is closely linked to hypothalamic and autonomic stress centres[124], and sleep stage transition (that is, from NREM sleep to wakefulness).

Finally, a large body of evidence suggests that sex differences exist at the molecular, cellular and systems levels of pain processing, which explains the female predominance in chronic pain disorders[125]. Given that pain is strongly linked to sleep disruption, sexual dimorphism in sleep quality might be based more on pain in females than in males. A study in osteoarthritis found associations between greater insomnia severity and lower pressure pain thresholds for non-Hispanic white patients but not for African American patients[126].

Sleep and pain

The relationship between sleep and pain is reciprocal; daytime pain predicts poor sleep and in turn, poor sleep predicts elevations in pain the following day[127]. Moreover, self-reported poor sleep confers a two-to-threefold risk of developing new onset chronic pain, especially in females[128], and predicts emergence, progression and persistence of musculoskeletal pain[128,129]. Poor sleep also increases the risk of transitioning from acute to chronic pain, along with progression of localized regional pain to widespread pain[130].

Sleep difficulties and pain often cluster together with depressed mood in patients with RA[131]. Insomnia is not only comorbid with

Table 1 | Quantitative sensory testing measures of pain sensitivity and central sensitization

Index	Assessment	Effects of experimental sleep loss/ disruption
Generalized hyperalgesia	Sensory testing extra-articular anatomical site	Not known
Heat pain threshold	Thermal stimulator with slow ramp in temperature from non-noxious warm temperature to noxious heat. Person indicates when stimuli (temperature) first feels painful	↓; Mechanism might include elevations in cellular inflammation (TNF and IL-6)
Pressure pain threshold	Algometer applied to muscle belly at a constantly increasing rate of pressure. Person indicates when stimuli (applied pressure) first feels painful	↓; Sleep-related mechanisms unknown
Temporal summation	Multiple (typically 10) brief (0.5-s) mechanical or thermal painful stimulations of equal intensity with a brief (2.5-s) constant inter-stimulus interval. Persons rate the intensity of the first stimulation and the peak intensity. The temporal summation index is expressed as a windup ratio, i.e. peak rating/first rating or as a difference score (peak-last)	↑; Mechanisms involve upregulation of second-order dorsal horn neurons but sleep-related mechanisms are unclear. Females may be differentially impacted
Conditioned pain modulation	Involves a phasic measure of pain sensitivity (e.g. pressure-pain threshold at the trapezius) and a tonic pain conditioning stimulus (e.g. 20-s cold pressor task) applied to a distal contralateral anatomical region from phasic stimuli. The phasic measure of pain sensitivity is assessed before and immediately after the conditioning stimuli. Conditioned pain modulation is indexed as the change (percentage change or difference score) in sensitivity before and after the conditioning stimuli. Under normal circumstances, pain sensitivity is reduced by the condition stimuli	↓; Sleep disruption impairs opioidergic analgesic systems but the mechanisms are unclear. Females are differentially impacted

depressive disorder, but also serves as a potent risk factor for depression[132,133]. In turn, depression, including a past history of depressive disorder, is associated with greater pain overall and with more stress-reactive pain in patients with RA[134], with evidence that depression predicts increases in symptoms of pain, which in turn, worsen insomnia complaints and further increase reports of depression[135].

Few studies have evaluated the unique relationship between sleep and pain in patients with RA. One prospective study found that baseline sleep problems predicted heightened RA pain severity a year later[136]. In addition, a single night of experimental partial sleep loss was associated with heightened pain, fatigue and depression in RA compared with responses in controls[52]: patients with RA showed exaggerated increases in self-reported pain symptoms, increases in the number of painful joints and severity of associated joint pain, and increases in clinician-rated painful and tender joints compared with responses in controls[52]. Patients with RA also showed exaggerated increases in symptoms of depression, yet depression did not mediate increases in pain responses[52].

Pain in RA is shaped by a complex interaction between joint damage, peripheral sensitization of nociceptive afferents by inflammatory mediators (such as cytokines and other sensitizing molecules) and central pain-processing mechanisms involving bottom-up and top-down modulation of pain signals.

The mechanisms linking sleep disruption to pain are poorly understood. Even in patients with well-controlled RA, pain is often prevalent; this observation suggests a central sensitization component, which might be induced by sleep disturbance[137]. Central sensitization involves a reorganization of spinal cord circuitry and supraspinal neural pathways that modulate noxious peripheral input with dysregulated central pain-inhibitory and -facilitatory mechanisms that amplify and sustain pain[138]. Consistent with central sensitization, patients with RA display generalized hyperalgesia at extra-articular anatomical sites[139], which is thought to be a consequence of peripheral, spinal and CNS alterations[140].

Several pain measures, including heat pain threshold, pressure pain threshold, temporal summation and conditioned pain modulation (CPM) are used to quantify pain sensitivity and modulation (Table 1). Quantitative sensory testing measures of pain threshold can be applied to disease active joints to index peripheral sensitization and at unaffected sites to index generalized hyperalgesia (that is, heightened central pain processing). Dynamic quantitative sensory testing measures of pain include temporal summation and CPM, which indirectly measure central pain-facilitatory processes and descending pain-inhibitory capacity, respectively. Temporal summation, a common central sensitization measure of pain facilitatory processing, involves heightened pain responses to repetitive noxious stimulation of the same intensity, which predicts development of chronic pain[141]. Notably, the NMDA receptor is upregulated by sleep deprivation. Further, a deficit in CPM also predicts the development and trajectory of chronic pain[142], and enhanced clinical pain[143], as observed in chronic pain conditions (such as osteoarthritis)[144]. CPM can be modulated by monoaminergic and opioid-mediated supraspinal mechanisms[145].

In RA, emerging evidence suggests that aberrant temporal summation and CPM processes might contribute to pain. Elevated temporal summation was found in patients with RA with active disease[146], which correlated with RA disease activity[147] as well as increased clinical pain intensity[148]. Enhanced baseline temporal summation was also found to mediate the relationship between sleep and pain in RA in patients receiving DMARDs, suggesting that temporal summation might contribute to pain reports in patients with RA who have low levels of localized joint inflammation[149]. Deficient CPM is also reported in patients with RA[150,151], and low CPM predicts poorer response to DMARD therapy[152] and increases in tender joint counts[147].

Experimental sleep disruption induces generalized hyperalgesia (decreased pressure pain and heat pain thresholds) in healthy volunteers without RA, and leads to central sensitization-enhanced temporal summation of mechanical pain, especially in females[153], and increased temporal summation to cold stimuli[154]. Moreover, sleep deprivation also impairs CPM and pain inhibitory capacity in females[155–158], but not in males[156]. Observational findings indicate that poor sleep correlates with reduced CPM in several musculoskeletal pain populations[126,159]. Further, poor sleep quality partially mediates the relationship between CPM and RA, in which baseline CPM significantly accounts for the relationship between poor sleep and clinical pain[147]. These data, coupled with experimental disruption studies, indicate that sleep quality might drive clinical pain reports in RA by impairing endogenous pain

inhibitory capacity, even in patients taking DMARDs[149]. Interventions designed to enhance sleep might therefore have a role in long-term pain management. Together, multiple pathways, including central sensitization, have a role in the relationship between sleep disturbance and pain in RA.

Sleep, pain and inflammation

Experimental research has found that sleep loss leads to aberrant increases in monocyte production of IL-6 and TNF[94], and that inflammation can heighten pain sensitivity[160–162]. However, research evaluating whether inflammation mediates the link between sleep disturbance and pain is limited, especially in humans. Moreover, as noted above, disturbances in sleep architecture such as loss of slow wave or N3 sleep seem to be uniquely associated with increases in inflammation[11,55], as well as with pain responses[163–165]. Furthermore, loss of N3 sleep, but not REM sleep, is associated with increases in markers of inflammation[11,55]. We interrogated the separate and joint contributions of N3 sleep and inflammation on pain sensitivity in healthy adults without RA. Using a multilevel approach, the effects of sleep disruption on hyperalgesia (heat pain threshold) were indirectly mediated by decreases in N3 sleep, which in turn led to increases in inflammation, which together served as double mediators to drive more than one-third of the pathway linking sleep disruption and pain sensitivity. Importantly, despite the effects of sleep disruption on total sleep time and REM sleep, neither of these sleep measures mediated changes in pain sensitivity[166]. Another experimental study found that sleep restriction led to elevations of circulating levels of IL-6, which in turn predicted elevated serum levels of IL-6, which in turn predicted increases in subjective reports of pain in healthy volunteers[167]. To the extent that loss of N3 sleep occurs in patients with RA who are experiencing sleep disruption, risk of heightened inflammation and pain sensitivity might be mitigated by interventions that target N3 sleep. The implications of such research would be substantial given the ubiquitous pattern of sleep fragmentation that occurs in association with pain and in patients with RA[34,35,168].

Several mechanisms might contribute to sleep- and inflammation-related hyperalgesia. Proinflammatory cytokines have receptor-mediated sensitizing effects on afferent nociceptive pathways[169,170] that mediate muscle and joint hyperalgesia[171,172], and sensitize nociceptors in peripheral nerve terminals[173,174]. Additionally, sleep disruption heightens SNS activity[11,116,175], which increases activity of NF-κB[99], upregulates inflammatory transcriptional profiles and increases expression of inflammatory mediators[176,177]. Alternatively, increases in the parasympathetic nervous system (that is, a shift towards vagal dominance) are associated with decreases of inflammation markers, with attendant benefit in animal models of arthritis[178,179]. In patients with RA, lower vagal activity is found along with increases in inflammation[180]. However, the potential links between sleep, parasympathetic activity and pain have not been evaluated.

Interventions for insomnia

Given that sleep disturbance leads to increases in inflammation and pain sensitivity, and both inflammation and pain can have reciprocal effects to induce further disturbance of sleep in a feedforward loop, an urgent need exists to evaluate the efficacy of interventions that target insomnia and interrupt this cycle.

Sedative hypnotic medications are often used to treat insomnia. A systematic review of pharmacotherapy for insomnia identified only six studies (all from before 2000) testing the short-term (1 night to 4 weeks) effects of benzodiazepines, benzodiazepine receptor agonists, zopiclone and eszopiclone in RA[181]. Substantial heterogeneity precludes a formal meta-analysis, but improvements in patient-reported sleep continuity outcomes were observed in most studies, as confirmed in a controlled trial with eszopiclone[182]. However, benzodiazepine receptor agonists are not typically recommended for long-term use and can increase the risk of falls in older adults. Furthermore, given advances in DMARDs, which have powerful effects on inflammation and disease activity, there is a need to study the effects of various sedative hypnotics on sleep and disease activity in patients receiving these newer biologic agents. To date, new classes of sedative hypnotics, including melatonin receptor agonists and dual orexin receptor antagonists, are also yet to undergo systematic scientific investigation in RA. In addition, pharmacotherapy poses risks for daytime impairments in cognitive function and depressed mood, and can lead to physiological dependency and only temporary improvement in insomnia.

Sleep education therapy, a behavioural programme, is widely used to target day-to-day behavioural and environmental factors that contribute to poor sleep. Although this programme can improve sleep quality, these benefits are often not sustained to achieve remission of insomnia[183,184]. Cognitive behavioural therapy for insomnia (CBT-I) is currently recommended as the first-line treatment for insomnia; CBT-I is a multi-component non-pharmacological treatment that combines cognitive therapy, stimulus control, sleep restriction, sleep hygiene and relaxation[185–187], with demonstrated efficacy to produce sustained remission of insomnia for up to 3 years[188]. Evidence also suggests that mindfulness-based treatments (such as mindfulness meditation and tai-chi) are non-inferior in the treatment of insomnia[189,190], with durable maintenance of benefit >1 year[189–191]. Interestingly – given the links between sleep disturbance, inflammation and pain sensitivity – CBT and pharmacotherapy are reported to improve sleep and pain symptoms in various populations of patients with chronic pain[192–196].

Despite the robust efficacy of CBT-I in adults, older adults[197] and in those with chronic pain[198], studies in RA are limited, and have not systematically reported or examined whether pharmacological treatments directed at inflammatory pathways (that is, DMARDs) moderate these effects. One trial found that CBT-I improved patient-reported insomnia symptoms and standard self-reported sleep continuity parameters[199], but not PSG measures. However, CBT-I has been reported to improve insomnia symptoms and PSG measures in patients with osteoarthritis[200].

Given the evidence that exercise and mindfulness-based interventions can improve insomnia in adult populations, randomized controlled trials have begun to evaluate the efficacy of these approaches in patients with RA. One study found that intermittent aerobic exercise was associated with a significant improvement in PSG measure of sleep efficiency, but not in usual care control; self-reported measures of sleep continuity and sleep quality did not improve in either group[201]. Furthermore, a 28-session walking-based intervention over 8 weeks improved sleep quality more than an education control intervention in patients with RA, the majority of whom were taking DMARDS (75%)[202]. Finally, delivery of a progressive muscle relaxation programme improved sleep quality and fatigue more than usual care in patients with RA[203], although a pilot study of relaxation-based yoga had no effect on self-reported symptoms of pain or sleep in a small sample of patients with RA, the majority of whom were being treated with DMARDs (92%)[204]. Two of the trials above enrolled patients with clinically significant insomnia symptoms despite the predominant use of DMARDs. Even in patients receiving DMARDs, symptoms of insomnia are prominent; adjunctive treatments of insomnia might therefore be of benefit[202].

Glossary

C-reactive protein
(CRP). An acute phase protein that is synthesized by the liver in response to the production of IL-6 by macrophages or T cells.

Electroencephalography
(EEG). A measure that tracks the electrical activity of the brain; one use is to graphically represent stages of sleep, which are defined by differences in waveform shape, frequency and amplitude.

Electromyography
A measure that tracks the electrical activity of muscle; it can be used, together with the EEG, to define stages of sleep, such as rapid eye movement sleep, in which low muscle tone or activity is accompanied by random and rapid eye movements.

Epigenetic ageing
An estimate of biological age given by evaluating changes in DNA methylation at particular genomic locations, which is found to be more predictive of mortality risk than chronological age.

Experimental sleep disruption
Imposing a loss of sleep during the night, for either part of the night (in other words, partial night sleep deprivation) or for the entire night (in other words, total night sleep deprivation).

Glucocorticoid resistance
A state of decreased sensitivity to the anti-inflammatory effects of glucocorticoids, which can be caused by ongoing increases in inflammation as well as by a genetic predisposition.

Glucocorticoids
Neuroendocrine hormones that belong to the steroid hormone class, which suppress inflammation and antiviral immune responses, in addition to having a role in the metabolism of protein, fat and glucose.

Hypothalamic–pituitary–adrenal axis
(HPA axis). A neuroendocrine system that links the hypothalamus, pituitary and adrenal glands and functions to regulate the immune system in response to circadian signalling, behavioural states such as sleep and peripheral inflammatory signals.

Parasympathetic nervous system
A component of the autonomic nervous system that comprises nerve fibres that innervate visceral tissues to regulate actions of the body when it is at rest, mainly through the release of the neurotransmitter acetylcholine.

Rapid eye movement
(REM). A stage of sleep, also known as paradoxical sleep, that is characterized by desynchronized electroencephalogram activity in a manner similar to waking, accompanied by random and rapid movement of the eyes together with low muscle tone. REM sleep is viewed as the sleep period in which there is a propensity to dream.

REM density
A measure specific to REM sleep and refers to the number of eye movements during REM sleep, which increases throughout the night along with a reduction in the drive to sleep. In other words, REM density is higher during the circadian or sleep period with arousal, and decreased in the night following sleep deprivation, which increases the drive to sleep.

Sleep continuity
The relative distribution of uninterrupted sleep, as opposed to wakefulness, during the night, as measured by sleep efficiency and wake time after sleep onset (the amount of time spent awake after turning off the lights and initiating sleep).

Sleep duration
The amount of time spent asleep during the night, measured either by subjective report or objectively, using polysomnography or actigraphy. Short sleep duration is defined as less than the reference amount of 7 h per night and is typically characterized as being less than 6 h of sleep per night. Long sleep duration is typically characterized as being more than 8 h of sleep per night.

Sleep efficiency
Time spent asleep as a percentage of the total time spent in bed.

Sympathetic nervous system
(SNS). A component of the autonomic nervous system that comprises nerve fibres that innervate lymphoid tissues, as well as nearly all other body tissues. The SNS regulates immune cell traffic and immune responses during sleep and in response to stress through the release of noradrenaline.

Given the experimental and longitudinal literature demonstrating a reciprocal relationship between pain and sleep, both the pharmacological and behavioural clinical trials literature have explored whether improving sleep translates into improved pain outcomes in RA. For example, pharmacotherapies are reported to improve pain[181], morning stiffness[196] and self-reported disease measures[199] in patients with RA and insomnia. Meta-analytic observations demonstrate that CBT-I improves clinical pain in non-RA samples[198,205] with chronic pain.

Finally, although evidence indicates that sleep disturbance can persist despite treatment with DMARDs[89], inflammatory cytokines are well recognized to alter sleep[75,206]. Given that these cytokines have a role in RA, further research is necessary to determine whether neutralization of these cytokines could effectively treat insomnia and associated pain symptoms in RA.

Insomnia treatment and reversal of inflammation

In addition to the efficacy of CBT-I and other behavioural interventions for treating insomnia, evidence indicates that insomnia-related inflammatory activation profiles are reversed in association with insomnia remission, although findings are largely based on populations without chronic inflammatory disease such as RA. For example, in older adults, clinical trial data show that insomnia interventions (including CBT-I or mindfulness-based treatments) lead to the following: decreases in sympathetic activity (which is known to activate inflammatory mechanisms[207–209]); decreases in levels of systemic inflammation including CRP levels; decreases in cellular inflammation such as stimulated monocyte production of pro-inflammatory cytokines; and a reversal of the inflammatory transcriptional profile[210–212]. The effect of insomnia treatment on inflammation might be clinically important, as CBT-I leads to a decrease in the proportion of patients with insomnia who have elevated levels of CRP (as defined by the high risk threshold of >3 mg/dl CRP); the magnitude of this anti-inflammatory benefit is comparable with that found with vigorous physical activity[213] or weight loss[214]. Interestingly, patients with insomnia who received a mindfulness-based treatment demonstrated robust decreases in TLR4-stimulated monocyte production of proinflammatory cytokines and downregulation of inflammatory transcriptional profiles[212] even in the absence of complete remission of insomnia. These results suggest that the ability of these treatments to reduce stress arousal and related sympathetic activation might have additional physiological

benefits separate from those driven by insomnia[215,216]. Some data also indicate that behavioural interventions that target depression and sleep complaints might extend to patients with RA. For example, CBT for depression improves depressive symptoms and reduces stimulated production of IL-6 in patients with RA[217].

The HPA axis has an important role in normal sleep in which CRH increases light sleep and wake time after disease onset[218]. Low doses of glucocorticoids increase deep sleep and vice versa[219]. A situation with a chronic inflammatory load such as in patients with RA might support this problem. Patients with RA and high serum cortisol levels demonstrate less sleep efficiency and somewhat later wake time after sleep onset. However, anti-TNF therapy (etanercept) completely reverts this interrelation without apparent effects on cortisol levels[220]: patients demonstrated longer total sleep time, better sleep efficiency and shorter awake time after sleep onset[68]. In other words, anti-TNF therapy changed the influence of cortisol on sleep parameters in RA, even though levels of cortisol did not change.

Conclusions

The CNS and the immune system reciprocally interact such that sleep enhances immune defences during the night, and in turn resting levels of cellular inflammation immediately prior to sleep can promote sleep continuity and depth in healthy adults. Sleep is a metabolically quiescent period, which means that energy sources are available during sleep to support innate as well as antiviral immune responses, which have a high metabolic demand. However, when activation of innate immune responses becomes chronic, for example, owing to an inflammatory disease such as RA, the homeostatic crosstalk between sleep and immunity becomes mis-aligned. For example, in patients with RA and co-morbidities, levels of inflammation become elevated, and these excessive inflammatory afferent signals lead to a disruption of sleep, which is further amplified by autonomic arousal mechanisms that are also activated in RA. Thus, sleep disruption activates SNS outflow, which steers the immune system towards greater increases in inflammation, which in turn magnifies sleep disruption in a positive feedback spiral. Maladaptive responses of the neuroendocrine system (for example, downregulation of glucocorticoid receptor sensitivity) further sustain this imbalance; in this setting, the inflammatory response is no longer effectively inhibited by cortisol. In RA, sleep disruption coupled with elevated inflammation induces hyperalgesia, increases pain sensitivity, and leads to progression of disease activity. The ability of pharmacological and behavioural interventions to treat insomnia and reverse insomnia-related inflammation reveals the potential to redirect mis-aligned inflammation. In other words, patients with RA might be able to accommodate the physiological insult of their inflammatory disorder more effectively if insomnia is treated.

To understand how distinct differences in RA map onto the risk of sleep disturbance and related immunological signatures, research on the environmental factors, medical co-morbidities, and psychosocial inputs that influence sleep and inflammatory biological mechanisms is needed. Multiple factors affect sleep quality, yet research in patients with RA remains limited. Improved understanding of the risk profile of sleep disturbance, including inflammatory and genomic differences, would substantially advance efforts to prevent the emergence of insomnia in RA, and provide insight into what specific aspects might be targeted to augment the therapeutic response of insomnia treatments.

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